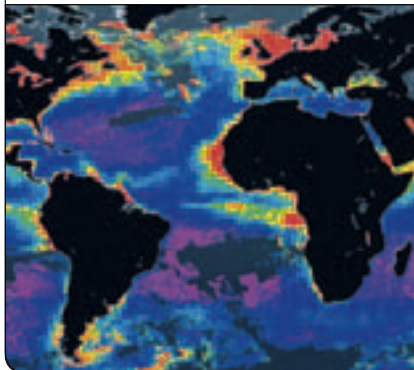


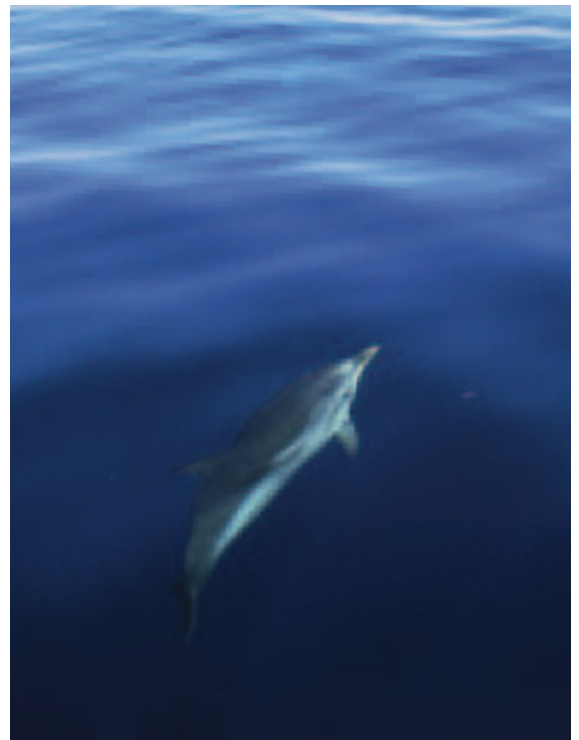
THE OPEN OCEANS

In deep oceans, the nutrients that fuel plankton growth settle on the ocean floor far below the sunlit zone. This means that there is often very little plankton, especially in deep tropical oceans where the warm surface water rarely mixes with deeper, richer water. In cooler oceans, seasonal storms stir nutrients into the surface waters, allowing marine life to flourish. Yet, even in the tropics, local upwelling can create dense concentrations of ocean life.

MIXING ZONES



This satellite image of plankton distribution shows the barren zones of tropical oceans in pink and deep blue. They contrast with vivid areas of orange, yellow, and green indicating the dense plankton of cold oceans and upwelling zones. Here, nutrient-rich water is brought to the surface by currents, and in cold oceans by seasonal storms and the sinking of chilled surface water. This causes plankton blooms in cold oceans in the fall and spring.



▲ CLEAR BLUE WATER

In open tropical oceans, sun-warmed surface water forms a floating layer above deep, cold water containing nutrients. The warm and cold water do not mix and this cuts off the nutrient supply that is vital to life in the sunlit zone, resulting in very little plankton. The water is crystal clear and virtually lifeless.

► NOMADS AND MIGRANTS

Plankton is widely scattered in tropical oceans, and so are the animals that eat it. Many, such as big manta rays and whale sharks, wander over vast areas in search of local concentrations of food. In cold oceans, the plankton blooms of fall and spring encourage more regular migrations. Shoaling fish, dolphins, and giants like this humpback whale arrive at the same times each year to make the most of the seasonal riches and then leave when the food runs out.

open oceans

